THE FEDERAL POLYTECHNIC MUBI
BIOLOGICAL SCIENCE TECHNOLOGY

COURSE: Plant Pathology
CODE: STB 315/HND 425
CLASS: HND 1 Biol,
HND 1 Biol./Microbiology
HND 2 Env. Bio
LECTURER: Ben Toy

LEARNING OBJECTIVES

At the end of this class, the students should be able to

- Explain what "Non-Infectious Diseases" are.
- Explain the principles of non-infectious diseases attacks.
- Explain the three classes of "non-infectious diseases".
    - Weather Conditions
    - Nutritional Conditions
    - Pollution
- Outline and explain the weather elements that cause diseases in plants.
- Outlined and explain the nutritional elements that cause disease conditions.
- Outline and explain the various pollutants that cause diseases in plants.
- Explain how non-infectious diseases can be controlled.

NON-INFECTIOUS PLANT DISEASE (ENVIRONMENTAL CONDITIONS)

- All plants have optimum environmental conditions at which they perform best.
- The environmental conditions include temperature, moisture, soil nutrients, light, PH etc.

Diseases caused by these conditions are called non-infectious diseases. These diseases are caused by the lack or excess of any of the above-mentioned conditions that support life.

- The non-infectious disease cannot be transmitted from a diseased plant to a healthy one
- They affect plants in all stages of growth such as seeds, seedlings, mature plants or fruits.
- They may cause damage in the field storage or market.

PRINCIPLES OF NON-INFECTIOUS DISEASES ATTACK

- The extremes of environmental conditions/Nutrients directly affect the physiological activities of the plant, making the plant to be diseased.
- The various pollutants act as "poison" and attack various organs of the plants (just as pathogens do) and cause diseases in the plants
- In conjunction with the pathogens that cause diseases by weakening the plant and making it easier for the pathogen to attack
- They also provide favorable conditions for the growth and development of the pathogen.

CLASSIFICATION OF NON-INFECTIOUS DISEASES

- In this course, we are going to study Non-Infectious Diseases under three broad classes

i. Weather conditions.
ii. Nutritional Conditions
iii. Pollution

WEATHER CONDITIONS

- The weather factors that most seriously affect the initiation and development of infectious plant diseases are temperature, moisture, light, soil nutrient, and soil PH.

Temperature

- Plant pathogens differ in their preference for high or low temperatures.
- While some pathogens develop best in areas and seasons or years with cooler temperatures, some others prefer relatively high temperatures.
- Some species of fungi e.g., *Typhula* and *Fusarium* which cause snow mould of cereals thrive only in cool temperatures.
- Also, the late blight pathogen *Phytophthora infestans* is most serious in northern latitudes and sub-tropics it is serious only during winter.
- On the other hand, most diseases are favoured by relatively high temperatures
    - e.g., the fusarium wilt of plants,
    - the brown rots of stone fruits,
    - southern bacterial rust of solanaceous plants caused by *Pseudomonas, solonacaerum* etc.

Moisture

- Moisture disturbances on plants cause harm to plants more than any other single environmental factor. If soil water is more than required in an area, it will lead to reduced growth, diseased appearance and even death of plants.
- The moisture may exist as rain or irrigation water on the plant surface or around the roots, as relative humidity in the air and as dew.
- Moisture mostly affects the germination of spores and the penetration of the host by the pathogens.
- Moisture such as splashing rain and running water also plays an important role in the distribution and spread of many of these pathogens.
- Moisture also affects disease by increasing the succulence of the host plant and thus increasing its susceptibility to certain pathogens.

Wind

- Wind affects plant diseases mostly by spreading the pathogen and to a lesser extent by drying up the moisture available to the pathogen.
- Most diseases spread rapidly because epidemics are caused by pathogens that are readily distributed by wind.
- Wind may cause injuries to plants while they are blown about and rubbing against each other; this facilitates infections by many pathogens as it creates entry points for them.

Soil pH

- The pathogens mostly affected by PH are the soil-based pathogens e.g., the club root of crucifers caused by *Plasmodiophora brasicae* is most prevalent drops sharply between 5.8 and 6.2 and is completely checked at 7.8.

Host plant nutrition

- When the plant is well nourished, it stands a higher chance of fighting diseases because its immunity is high,
- But with poor nutrients, the plants are more predisposed to diseases as the immunity of the host plant is low.

Light

- Light plays a very vital role in plant nutrition (photosynthesis)
- Lack of sufficient light retards chlorophyll formation and hence reduces the rate of photosynthesis
- This will result in the production of pale green leaves, slender growth with long internodes, a premature drop of leaves and flowers (etiolating) etc.

Oxygen

- Low oxygen conditions are always due to water logging and high temperature
- This may cause damage to the roots of plants, especially in water-logged areas.

POLLUTION

- Almost all air pollutions causing plant injury are gases, but some particulate matter or dust also affects vegetation.
- Table 1 below shows air pollutants, sources and effects on plants.

NUTRIENT CONDITION

- The kind of symptoms produced by a deficiency of a certain nutrient depends primarily on the functions of that particular element in the plant.
- These functions are inhibited or interfered with when the element is limiting.
- The table 2 below shows some elements, their functions, and the symptoms produced in plants when they are deficient.

CONTROLLING NON-INFECTIOUS DISEASES

- Noninfectious plant disease can be controlled by:
    - Avoiding the extremes of the environmental conditions responsible for such diseases.
    - By protecting the plants with substances that will bring these conditions to moderate levels.
    - Applying the correct/right amounts of fertilizers at the right time to avoid “mal-nutrition”
    - Avoid highly polluted sites.
    - The right planting season and time must to known and done properly.

Table 1: Pollutants and their Effects on Plants

Pollutant	Sources	Susceptible plants	Symptoms	Remark
Ozone (0_3)	Exhaust, (Released NO3+ atmospheric O2 I sunlight – O3) from stratosphere light running and forest	Expanding leaves of all plants, especially tobacco beans, cereals, alfalfa, pine, citrus corn etc.	Chlorosis of leaves, spots are small to large, bleached white to tom, Premature defoliation and stunting occur in plants such as citrus, grapes and vines.	Enters through stomata it is the most destructive air pollutant to plants
Dust	Roads, cement factories, burning local etc.	All plants	Form dust or crusty layers on plant surfaces. Plants become chlorotic, grow poorly and may die. Some dust is toxic and burns leaf tissues directly or after dissolving in dew or rainwater	
Ethylene	Exhaust, burning gas fuel, oil and coal. From ripening fruit in storage	Many kinds of plants toxic at 0.05ppm	Plants remain stunted, their leaves develop abnormally and senescence prematurely. Plants produce fewer blossoms and fruit, e.g. apples, and develop depressed necrotic bark areas.	Ethylene is a plant hormone with numerous functions
Chlorine (Co) and Hydrogen chloride (HCl)	Refineries glass factories and incineration of plastics	Many kinds of plants are usually near the sources. Toxic at 0.1ppm	Leaves show bleached necrotic areas between veins, leaf margins often appear scorched. leaves may drop prematurely – damage resembles that caused by SO_2	
Hydrogen fluoride (HF)	Stacks of factories processing ore or oil	Many kinds of plants, including corn, and peach, especially wet leaves are most sensitive. toxic at 0.1 – 0.2 ppm	Leaf margins of cheats and leaf tips of monocots twin tan to dark brown, die and may fall from the leaf. Some plants tolerate HF up to 200 ppm	HF may evaporate or be washed out of the plant and the plant recovers slowly.
Peroxyacyl nitrate (PAN)	Exhaust gas	Many kinds of plants include spinach, petunia, tomato, lettuce etc.	Causes “silver leaf” on plants ie bleached white to brown spots on the lower surface of leaves that may later spread throughout leaf thickness and resemble ozone injury	Particularly since near metropolitan areas with 5mog and inversion layers
Sulfur dioxide (SO_2	Stacks of factories exhausts and other internal combustion engines	Many kinds of plants including alfalfa, conifers, peas, cotton, bean etc. toxic at 0.3 to 0.5 ppm	Low concentration causes general chlorosis, Higher concentrations cause bleaching of inter-viral tissue of leaves.	It also combines with moisture and forms toxic acid droplets (acid rain)
Nitrogen dioxide (NO_2)	From oxygen and nitrogen in the air by hos, combustion sources e.g. furnaces etc.	Many kinds of plants especially, beans, tomatoes etc. toxic at 2 to 3 ppm	Causes bleaching and browsing of plants similar to that caused by SO_2 at low concentrations. It also suppresses the growth of plants.	

Table 2: Plants Nutrients and their Deficiency Effects

Nutrient	Function	Symptoms when deficient
Nitrogen N	Present in most substances of cells	Plants grow poorly and are light green the lower leaves turn yellow or light brown and the stems are short and slender.
Phosphorus P	Present in DNA, RNA phosphorus (membranes) ADP, ATP, etc.	Plants grow poorly and the leaves are bluish-green with purple tints. Lower leaves sometimes turn light brown with purple spots. Shoots are short and thin, upright and spindly
Potassium K	Acts as a catalyst for many reactions	Plants have thin shoots which in severe cases, show dieback. Older, leaves show chlorosis with browning of the tips. Scorching of the margins and many brain spots are usually near the margins. Fleshy tissue shows end necrosis.
Magnesium Mg	Present in chlorophyll and is part of many enzymes	First, the older, then the younger leaves become mottled or chlorotic, then reddish, and sometimes necrotic spots appear. The tips and margins of leaves may twin up words and the leaves appear cupped. Leaves may drop off.
Calcium Ca	Regulates the permeability of membranes. Affects the activity of many enzymes	Young leaves become distorted, with their tips hooked back and the margin curled. Leaves may be irregular in shape and rugged with brown scorching spots, and terminal buds finally die. The plants have poor, bare root systems. Causes blossom end rot of many fruits.
Boron B	Affects the translocation of sugars and utilization of calcium in cell wall formation	The basis of young leaves of terminal buds becomes light green and finally breaks down. Stems and leaves become distorted. Fruit, fleshy roots or stem etc., may crack on the surface and/or rot in the Centre. Causes many plant diseases e.g. of turnips browning or hollow stem of cauliflower, cracked the stem of celery, corky spot, hard fruit of citrus etc.
Sulfur S	Present in some amino acids and coenzymes	Young leaves are pale green or light yellow without any spots. The symptoms resemble those of Nitrogen deficiency.
Iron Fe	It is a catalyst of chlorophyll synthesis. Part of many enzymes.	Young leaves become severely chlorotic, but their main veins remain characteristically green. Sometimes brown spots develop. Part of our entire leaves may dry. Leaves may be shed.
Zinc	Is part of enzymes valued in auxin synthesis and oxidation of sugars	Leaves show interveinal chlorosis. Later they become necrotic and show purple pigmentation. Leaves are few and small, inter nodes are short. From rosettes and fruit production is low. Leaves are short causing little leaves of apple's stone fruits and grapes, sickle leaf of cocoa, white tip of corn etc.
Copper Cu	Is part of many oxidative enzymes	Tips of young leaves of cereals wither and their margins become chlorotic leaves many fail to unroll and tend to appear milled. The heading is reduced and the heads are dwarfed and distorted. Citrus, pine and stone fruits show dieback of things in the summer, burning of leaf margins, chlorosis resetting etc. vegetable crops fail to grow.
Manganese Mn	Is part of many enzymes of respiration, photosynthesis and nitrogen utilization	Leaves become chlorotic but their smallest veins remain green and produce a checked effect. Necrotic sport may appear scattered on the leaf. Severely affected leaves turn brown and wither.